

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage: 40%

Read the given data carefully and compare the techniques or results based on multiple factors such as accuracy, robustness, and computational constraints. Evaluate and justify your decision with appropriate reasoning considering the application context.

By

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5. Motion Estimation vs Computational Cost

Two methods:

- **Method A: low error (5%), high computation**
- **Method B: moderate error (10%), low computation**

Compare the methods and evaluate their suitability for real-time vs offline applications.

Justify your decision.

Motion estimation is used to determine the movement of objects between consecutive video frames. The two methods differ mainly in **accuracy and computational cost**.

- **Method A:** It has a **5% error**, which means it provides highly accurate motion estimation. However, it requires **high computational power**, so processing is slower.
- **Method B:** It has a **10% error**, making it less accurate than Method A. But it has **low computational cost**, so it can process frames much faster.

Comparison

Factor	Method A	Method B
Error	5%	10%
Accuracy	High	Moderate
Computation	High	Low
Processing speed	Slower	Faster
Real-time use	Less suitable	More suitable
Offline use	More suitable	Suitable

Application Suitability

Real-time

applications:

Method B is preferred because applications such as **video surveillance, live video processing, video conferencing, and robotics** require fast processing. Although its error is 10%, the lower computational cost allows the system to respond quickly.

Offline

applications:

Method A is preferred because applications such as **video analysis, research, and post-processing** do not require immediate results. The system can spend more time calculating motion to achieve the lower 5% error.

Conclusion

Method A is best for offline applications where **accuracy is the main priority**. **Method B is best for real-time applications** where **speed and computational efficiency are more important**.

Therefore, the choice depends on the required balance between **motion estimation accuracy** and **processing speed**.

17. Stereo vs SfM Accuracy Comparison

Results:

Method	Accuracy	Hardware
Stereo	High	Dual camera
SfM	Moderate	Single camera

Compare and evaluate which is more practical under hardware limitations.

Stereo vision and Structure from Motion (SfM) are methods used to estimate **depth and 3D structure** from images.

Factor	Stereo	SfM
Accuracy	High	Moderate
Hardware	Dual camera	Single camera
Cost	Higher	Lower
Depth estimation	More accurate	Less accurate
Hardware requirement	More	Less
Practical under limitations	Less practical	More practical

Evaluation

Stereo:

Stereo vision uses **two cameras** to capture the same scene from different viewpoints. The difference between the two images is used to calculate depth. It provides **high accuracy** and reliable depth information. However, it requires a dual-camera setup, proper camera calibration, and synchronization, which increases hardware cost and complexity.

SfM:

Structure from Motion uses a **single camera** and multiple images taken from different viewpoints. It estimates the camera movement and reconstructs the 3D structure of the scene.

Its accuracy is **moderate**, but it requires only one camera, making it cheaper and easier to deploy.

Under Hardware Limitations

If hardware resources are limited, **SfM is more practical** because it requires only a single camera. Although its accuracy is lower than Stereo, the reduced hardware requirement makes it suitable for **mobile devices, low-cost systems, drones, and basic 3D reconstruction applications**.

Conclusion

Stereo → **Higher accuracy but requires more hardware.**

SfM → **Moderate accuracy but requires less hardware.**

Therefore, **SfM is the better choice under hardware limitations**, while Stereo should be selected when **high accuracy is more important than hardware cost and complexity**.

19. Motion Tracking Accuracy Comparison

Results:

Method	Accuracy	Stability
A	High	Low
B	Moderate	High

Compare and evaluate which is suitable for long-term tracking.

Motion tracking involves continuously detecting and following the movement of an object across multiple frames. The two methods differ in **accuracy and stability**.

Factor	Method A	Method B
Accuracy	High	Moderate
Stability	Low	High
Tracking precision	Better	Moderate
Long-term tracking	Less suitable	More suitable

Evaluation

Method **A:**

Method A provides **high accuracy**, so it can locate the object precisely. However, its **low stability** may cause tracking failures when the object changes position, becomes partially

hidden, or when there are changes in lighting. Therefore, it may not perform consistently over a long period.

Method

B:

Method B has **moderate accuracy**, but its **high stability** allows it to maintain tracking more reliably over many frames. Even if there is a small loss in precision, consistent tracking is more important for long-duration applications.

Suitability for Long-Term Tracking

For **long-term tracking**, **Method B is more suitable** because stability is essential for maintaining continuous tracking without frequent failures.

Examples include **surveillance systems, traffic monitoring, sports tracking, and video analytics**.

Conclusion

Method A → High accuracy but less stable.

Method B → Moderate accuracy but highly stable.

Therefore, **Method B is the better choice for long-term motion tracking**, because maintaining continuous and reliable tracking is more important than achieving maximum accuracy in every individual frame.

20. Surveillance Model Selection

Results:

	Model Accuracy	Latency
A	82%	30 ms
B	90%	70 ms
C	95%	120 ms

Compare and evaluate under strict latency constraints.

In a surveillance system, **accuracy and latency** are important. Accuracy determines how correctly the system detects objects or events, while latency determines how quickly it provides the result.

Model Accuracy Latency Evaluation

A 82% 30 ms Fastest, but lower accuracy

Model Accuracy Latency Evaluation

B	90%	70 ms	Good balance
C	95%	120 ms	Highest accuracy, but slowest

Evaluation

Model

A:

Model A has the **lowest latency of 30 ms**, making it highly suitable for strict real-time surveillance. However, its accuracy is only **82%**, so it may produce more incorrect detections.

Model

B:

Model B provides **90% accuracy** with **70 ms latency**. It offers a good balance between detection accuracy and response time, but may not be suitable if the latency limit is extremely strict.

Model

C:

Model C has the highest accuracy of **95%**, but its **120 ms latency** makes it slower. Under strict latency constraints, this delay may affect real-time surveillance performance.

Conclusion

If the surveillance system has a **very strict latency requirement**, **Model A is the best choice** because its 30 ms latency ensures fast response.

If the system can tolerate around **70 ms latency**, **Model B is preferable** because it provides significantly better accuracy while maintaining reasonable speed.

Model C should be avoided under strict latency constraints because its 120 ms latency is too high.

Strict latency → Model A

Balanced accuracy and speed → Model B

22. Depth Estimation Error Analysis

Results:

Method	Error
A	0.5 m
B	1.2 m
C	0.8 m

Compare and evaluate accuracy for real-world applications.

Depth estimation determines the distance between the camera and objects in a scene. **Lower error means higher accuracy.**

Method Error Accuracy

A **0.5 m Highest**

B 1.2 m Lowest

C 0.8 m Moderate

Evaluation

Method **A:**

Method A has the **lowest error of 0.5 m**, so it provides the most accurate depth estimation. It is suitable for applications where precise distance measurement is important.

Method **B:**

Method B has the **highest error of 1.2 m**, making it the least accurate. Such a large error can cause problems in applications that require reliable depth information.

Method **C:**

Method C has an error of **0.8 m**, which gives moderate accuracy. It can be used when extremely high precision is not required.

Real-World Applications

For applications such as **robot navigation, autonomous vehicles, obstacle detection, and 3D measurement**, accurate depth estimation is important. Therefore, **Method A is the best choice** because its 0.5 m error is the lowest.

Conclusion

Method A → Best accuracy (0.5 m)

Method C → Moderate accuracy (0.8 m)

Method B → Lowest accuracy (1.2 m)

Therefore, **Method A is most suitable for real-world applications** where accurate and reliable depth estimation is required.

25. Model Efficiency vs Accuracy

Results:

Model	Accuracy	Computation
A	90%	High
B	85%	Low

C 88% Moderate
Compare and evaluate for edge devices.

For edge devices, both **accuracy and computational cost** are important because devices such as cameras, mobile phones, and IoT systems have limited processing power, memory, and energy.

Model Accuracy Computation Evaluation

A	90%	High	Highest accuracy, but resource-intensive
B	85%	Low	Fast and efficient, but lower accuracy
C	88%	Moderate	Good balance between accuracy and efficiency

Evaluation

Model A:

Model A provides the **highest accuracy of 90%**, but it requires high computation. On edge devices, this can increase processing time, power consumption, and memory usage. Therefore, it may not be the most practical choice for resource-constrained devices.

Model B:

Model B has the **lowest computational requirement**, making it efficient and fast. However, its accuracy is only **85%**. It is suitable when power and processing resources are very limited and moderate accuracy is acceptable.

Model C:

Model C provides **88% accuracy with moderate computation**. It offers a good compromise between performance and resource usage, making it more practical for many edge-device applications.

Conclusion

For **strictly resource-limited edge devices**, **Model B** is suitable because of its low computational cost. However, if a balance between accuracy and efficiency is required, **Model C is the best choice**.

Model A → Highest accuracy but high computation

Model B → Most efficient but lower accuracy

Model C → Best overall balance for edge devices

27. Video Processing Pipeline Comparison

Results:

Pipeline	Accuracy	Speed
A	88%	Fast
B	92%	Moderate
C	95%	Slow

Compare and evaluate suitability.

A video processing pipeline should be selected based on the required **accuracy and processing speed**. Higher accuracy is useful for precise analysis, while higher speed is important for real-time applications.

Pipeline	Accuracy	Speed	Evaluation
A	88%	Fast	Best for real-time processing
B	92%	Moderate	Good balance
C	95%	Slow	Best accuracy, but slow

Evaluation

Pipeline

A:

Pipeline A provides **88% accuracy** and fast processing. It is suitable for **real-time applications** such as live surveillance and video streaming, where quick results are important.

Pipeline

B:

Pipeline B achieves **92% accuracy** with moderate speed. It provides a good balance between accuracy and processing time, making it suitable for applications that need both reasonable speed and reliable results.

Pipeline

C:

Pipeline C has the **highest accuracy of 95%**, but its slow processing makes it unsuitable for applications requiring immediate results. It is better for **offline video analysis**, where accuracy is more important than speed.

Conclusion

- Pipeline A → Real-time applications
- Pipeline B → Balanced applications
- Pipeline C → Offline/high-accuracy applications

Therefore, **Pipeline B is the most balanced choice**, while Pipeline A is preferred for strict real-time requirements and Pipeline C for applications where maximum accuracy is the priority.

30. System Performance under Constraints

Results:

Model	Accuracy	Latency	Cost
A	88%	50 ms	Low
B	92%	80 ms	Moderate
C	95%	120 ms	High

Compare and evaluate the best system considering all constraints.

A system should be selected by considering **accuracy, latency, and cost** together. A highly accurate model may not be practical if it has high latency and cost.

Model	Accuracy	Latency	Cost	Evaluation
A	88%	50 ms	Low	Fast and economical
B	92%	80 ms	Moderate	Good balance
C	95%	120 ms	High	Highest accuracy but expensive and slow

Evaluation

Model

A:

Model A has the lowest accuracy at **88%**, but it has the **lowest latency (50 ms)** and **low cost**. It is suitable for real-time systems with strict time and budget constraints.

Model

B:

Model B provides **92% accuracy** with **80 ms latency** and moderate cost. It offers a good compromise between accuracy, speed, and cost.

Model

C:

Model C achieves the highest accuracy of **95%**, but it has the **highest latency (120 ms)** and high cost. It is suitable only when accuracy is more important than speed and expense.

Conclusion

Considering **all three constraints**, **Model B is the best overall choice** because it provides a strong **92% accuracy** while maintaining reasonable latency and moderate cost.

- **Model A → Best for low cost and fast response**
- **Model B → Best overall balance** ✓
- **Model C → Best only when maximum accuracy is required**

Therefore, **Model B is the most practical system when accuracy, latency, and cost are all important.**